PROPOSITIONAL LOGIC

Proposition: A proposition or statement can be defined as a declarative sentence i.e, either T or F but not both.

* for eg: $3+3=6$
$3+3=7$ are both the statements where 1st is T & 2nd is F.

* eg: $x+y>1$ is not a statement because for some values of x & y the sentence is True where as for others it is false. The truth or falsity of a statement is called it's truth value. Since only 2 possible truth value are admitted, this logic sometimes called "2 valued logic".

* Note: Questions, Exclamations & commands are not proposition.

Ques: a) The Sun rises in the west False
b) $2+4=6$; ; True
c) $(5,6) \subset (7,6,5)$ True
d) Do you speak Hindi ?
e) $4-x=8$ Declarative
f) Close the door.
g) What a hot day ?
h) We shall have chicken for dinner. Proposition

Proposition

ATOMIC COMPOUND

1) ATOMIC PROPOSITION :- It is the individual synthe-tic element. A proposition consisting of only a single propositional variable or a single propositional constant is called atomic (primary, primitive) proposition i.e. they cannot be sub-divided, represented by letters small alphabets. eg: p, q, r, s.

2) COMPOUND PROPOSITION :- A proposition is obtained from the combination of 2 or more propositions by means of logical operator or connectors or two or more pro-position or by negating a propositions is referred as molecular or composite or comp-ound proposition.

eg: Passengers are waiting $ train come late
(S_1 = are waiting, S_2 = come late)

$S_1 \wedge S_2$

eg: Either Shreya or Akash is topper of class.

$S_1 \vee S_2$

Connectives in the Proposition Logic :-

Symbol Used	Connective Word	Nature of the compound statements formed by Connectives	Symbolic form	Negation
$\sim, \neg,], N$	not	Negation	$\sim P$	$\sim(\sim P) = P$
$\wedge$	and	Conjuction	$P \wedge Q$	$(\sim P) \vee (\sim Q)$
$\vee$	or	Disjunction	$P \vee Q$	$(\sim P) \wedge (\sim Q)$
$\rightarrow$	if---then	conditional	$P \rightarrow Q$	$P \wedge (\sim Q)$
$\leftrightarrow$	if and only if	Bi-conditional	$P \leftrightarrow Q$	$[P \wedge (\sim Q)] \vee [Q \wedge (\sim P)]$

1) Conjuction :- If P & Q are 2 statements, then conjunction of P & Q $(P \wedge Q)$ is the compound statement. Denoted by $(P \wedge Q)$ & read as "P and Q".

(1) eg: P : Rani is healthy.
Q : He has blue eyes.
$P \wedge Q$: Ram is healthy and has blue eyes.

(2) P : It is raining.
Q : It is cold
$P \wedge Q$: It is raining and cold.

Truth Table :-

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

2) Disjunction :- If P & Q are 2 statements, then disjunction of P & Q. Denoted by $(P \vee Q)$ & read as "P or Q".

(1) eg: P : Ram is healthy.
Q : He has blue eyes.
$P \vee Q$: Ram is healthy or has blue eyes.

Truth Table :

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

3) Conditional :- If P & Q are proposition, the compound proposition "if P then Q" denoted by "$P \rightarrow Q$" is called a conditional proposition.

$P \rightarrow Q$
↓ ↓
antecedent Consequence
(hypothesis) (conclusion)

eg: P : Tomorrow is Sunday. → is antecedent
Q : Today is Saturday → is consequence
$P \rightarrow Q$: If Tomorrow is Sunday then today is saturday.

Truth Table :

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

The connective if – then can be read as follows-

1) P is sufficient for q.
2) P only if q.
3) q is necessary for P.
4) q if P.
5) q follows from P.
6) q is the consequence of P.

4) Bi-conditional :- If P & q are statements then the compound statement "P if and only if q", denoted by $P \leftrightarrow q$ / $P \Leftrightarrow q$ / $P \rightleftharpoons q$. This can be stated as P is necessary & sufficient for q.

eg: $P \leftrightarrow q$: He swims if and only if water is warm
$P \leftrightarrow q$: He comes to college if and only if he wakes up early.

Truth Table :

P	q	$P \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

5) Negation :- If P is any proposition, the negation of P ($\sim P$), denoted by "$\sim P$" read as "not P" is a proposition which is false when P is true & true when P is false.

eg:- P : Ram is healthy.
$\sim P$: Ram is not healthy.

Truth Table :-

P	$\sim P$
T	F
F	T

Ques: (i) $P \wedge (\sim q \vee q)$
(ii) $\sim(P \vee q) \vee (\sim P \wedge \sim q)$

Soln (i)

P	q	$\sim q$	$\sim q \vee q$	$P \wedge (\sim q \vee q)$
T	T	F	T	T
T	F	T	T	T
F	T	F	T	F
F	F	T	T	F

} = P

(ii)

P	q	$\sim P$	$\sim q$	$P \vee q$	$\sim P \wedge \sim q$	$\sim P \vee q$	(ii)
T	T	F	F	T	F	F	F
T	F	F	T	T	F	F	F
F	T	T	F	T	F	F	F
F	F	T	T	F	T	T	T

Ques: $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

Soln:

P	Q	R	$Q \wedge R$	$P \vee (Q \wedge R)$	$P \vee Q$	$P \vee R$	$(P \vee Q) \wedge (P \vee R)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Ques: $P \rightarrow Q \equiv \sim Q \rightarrow \sim P$

Soln:

P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$	$\sim Q \rightarrow \sim P$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

$\equiv$

Ques: (I) $\neg(P \vee (\neg P \wedge Q)) \equiv \neg P \wedge \neg Q$

(II) $\neg(P \wedge Q) = \neg P \vee \neg Q$

(III) $\neg(P \vee Q) = \neg P \wedge \neg Q$

Soln: (I)

P	Q	$\sim P$	$\sim Q$	$\sim P \wedge Q$	$P \vee (\sim P \wedge Q)$	$\sim(P \vee (\sim P \wedge Q))$	$\sim P \wedge \sim Q$
T	T	F	F	F	T	F	F
T	F	F	T	F	T	F	F
F	T	T	F	T	T	F	F
F	F	T	T	F	F	T	T

$=$

★ $P \leftrightarrow q = (P \rightarrow q) \wedge (q \rightarrow P)$

★ $P \rightarrow q = (\sim P \vee q) \wedge (\sim q \vee P)$

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(II)

P	q	$\sim(P \wedge q)$	$\sim P$	$\sim q$	$\sim P \vee \sim q$
T	T	F	F	F	F
T	F	T	F	T	T
F	T	T	T	F	T
F	F	T	T	T	T

(III)

P	q	$\sim(P \vee q)$	$\sim P$	$\sim q$	$\sim P \wedge \sim q$
T	T	F	F	F	F
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

Algebra of Proposition :-

1) a) $P \vee P \equiv P$ b) $P \wedge P \equiv P$ → Idempotent law

2) a) $(P \vee q) \vee \gamma \equiv P \vee (q \vee \gamma)$ → Associative law

b) $(P \wedge q) \wedge \gamma \equiv P \wedge (q \wedge \gamma)$.

3) a) $P \vee q \equiv q \vee P$ → Commutative law

b) $P \wedge q \equiv q \wedge P$

4) a) $P \vee (q \wedge \gamma) \equiv (P \vee q) \wedge (P \vee \gamma)$ → Distributive law

b) $P \wedge (q \vee \gamma) \equiv (P \wedge q) \vee (P \wedge \gamma)$

5) a) $P \vee f \equiv P$

b) $P \wedge T \equiv P$ → Identity laws

6) a) $P \vee T \equiv T$

b) $P \wedge f \equiv F$

7) a) $P \vee \sim P \equiv T$

b) $P \wedge \sim P \equiv F$ → Complement laws

8) a) $\sim T \equiv F$

b) $\sim F \equiv T$

9) $\sim(\sim P) \cong P$ → Involution law

10) a) $\sim(P \vee q) \cong \sim P \wedge \sim q$
b) $\sim(P \wedge q) \cong \sim P \vee \sim q$
} → Demorgan's law

11) a) $P \wedge (P \vee q) = P$
b) $P \vee (P \wedge q) = P$
} → Absorption law

Some other Converse :-

If P & q are 2 Proposition then some other conditional propositions related to $P \to q$:

★ Converse : Converse of $P \to q$ is $q \to P$.

★ Contrapositive : Contrapositive of $P \to q$ is $\sim q \to \sim P$

★ Inverse : Inverse of $P \to q$ is $\sim P \to \sim q$.

Ques: $P \leftrightarrow q \equiv (P \vee q) \to (P \wedge q)$. By algebra of proposition.

Soln: R.H.S $(P \vee q) \to (P \wedge q)$

$\sim(P \vee q) \vee (P \wedge q)$

$(\sim P \wedge \sim q) \vee (P \wedge q)$

$(\sim P \wedge P) \vee (\sim P \wedge q) \vee (\sim q \wedge P) \vee (\sim q \wedge q)$

$F \vee (\sim P \wedge q) \vee (\sim q \wedge P) \vee F$

$(\sim P \wedge q) \vee (\sim q \wedge P)$

$\Rightarrow P \leftrightarrow q$. L.H.S Ans.

Ques: $\sim(P \vee (\sim P \wedge Q)) \cong \sim P \wedge \sim Q$.

Soln: LHS

$\sim(P \vee (\sim P \wedge Q))$

$\sim[(P \vee \sim P) \wedge (P \vee Q)]$

$\sim[T \wedge (P \vee Q)]$

$\sim(P \vee Q)$

$\Rightarrow \sim P \wedge \sim Q$. R.H.S Ans.

Ques (I) $P \rightarrow Q \cong \sim Q \rightarrow \sim P$

Ques (II) $\sim P \rightarrow \sim Q$ is logically equivalent to

i) $P \rightarrow Q$

ii) $Q \rightarrow P$

iii) $\sim Q \vee P$

iv) $\sim P \vee Q$.

Soln ii) $Q \rightarrow P$.

iii) $\sim Q \vee P$.

Soln (I) LHS

$P \rightarrow Q = P \wedge (\sim Q)$

Tautology:- A compound proposition that is always true for all possible truth value of it's variables of or in other words contain only T in the last column of it's truth table is called Tautology

Truth Table:

P	$\sim P$	$\sim P \vee P$
T	F	T
F	T	T

Contradiction:- A compound proposition that is always false for all possible values of it's variables or in other words contain only F in the last column of it's truth table is called Contradiction.

Truth Table:-

P	~P	P ∧ ~P
T	F	F
F	T	F

Contigency:- A proposition which is neither tautology nor Contradiction is called contigency.

Truth Table:-

P	Q	P ∧ Q
T	T	T
T	F	F
F	T	F
F	F	F

Satisfiability:- A compound proposition that obtains atleast one true value in the last/output column is called satisfiability.

Truth Table:

P	Q	P ∧ Q
T	T	T
T	F	F
F	T	F
F	F	F

Ques: Show that $(P \vee Q) \vee (\sim P \vee R) \rightarrow (Q \vee R)$. Given a statement either tautology, contradiction & contigency.

Sol^n: Let $S = (P \vee Q) \vee (\sim P \vee R) \rightarrow (Q \vee R)$

P	Q	R	~P	P∨Q	~P∨R	Q∨R	(P∨Q)∨(~P∨R)	S
T	T	T	F	T	T	T	T	T
T	T	F	F	T	F	T	T	T
T	F	T	F	T	T	T	T	T
T	F	F	F	T	F	F	T	F
F	T	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T	T
F	F	T	T	F	T	T	T	T
F	F	F	T	F	T	F	T	F

Hence, it is contigency Ans

Ques: Prove that $(\sim P \leftrightarrow \sim Q) \leftrightarrow (Q \leftrightarrow R)$ is tautology.

Sol^n: $(\sim P \leftrightarrow \sim Q) \leftrightarrow (Q \leftrightarrow R) \rightarrow$ Let S

P	Q	R	~P	~Q	~P↔~Q	Q↔R	S
T	T	T	F	F	T	T	T
T	T	F	F	F	T	F	F
T	F	T	F	T	F	F	T
T	F	F	F	T	F	T	F
F	T	T	T	F	F	T	F
F	T	F	T	F	F	F	T
F	F	T	T	T	T	F	F
F	F	F	T	T	T	T	T

Hence, it is not tautology. Ans

Ques: $\sim(P \wedge Q) \vee Q$

Sol^n: $(\sim P \vee (\sim Q) \vee Q)$

$\sim P \vee T \Rightarrow T$ It is tautology. Ans

Ques: $P \wedge (Q \wedge \sim P)$

Sol^n: $(P \wedge Q) \wedge (P \wedge \sim P)$

$(P \wedge Q) \wedge F = F$ It is contradiction not tautology. Ans.

Ques: $P \rightarrow (P \vee Q)$

Sol^n: $\sim P \vee (P \vee Q)$

$\Rightarrow (\sim P \vee P) \vee Q$

$T \vee Q$

T Tautology Ans

Laws of Conditional & Bi-conditional:-

★ $P \rightarrow Q \equiv \sim P \vee Q / P \wedge \sim Q$ (I)

★ $P \leftrightarrow Q \equiv (P \wedge Q) \vee (\sim P \wedge \sim Q)$ (II)

★ $P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$ (III)

By using (I) & (III)

★ $(P \leftrightarrow Q) = (\sim P \vee Q) \wedge (P \vee \sim Q)$

Ques: $(P \wedge Q) \rightarrow (P \vee Q)$

Sol^n: $\sim(P \wedge Q) \vee (P \vee Q)$

$\sim P \vee \sim Q \vee P \vee Q$

$\sim P \vee P \vee Q \vee \sim Q$

$\Rightarrow T$ It is tautology. Ans

Normal form :- The two compound Propositions 'P & Q' are said to be logically equivalent $P \equiv Q$ by means of some standard form is known as "normal form".

↳ There are 2 types of standard normal form. —

(1) Disjunctive Normal form (DNF)(SOP)
(2) Conjuctive Normal form (CNF)(POS)

(1) Disjunctive Normal form :- In an logical expression, a product of the variable & their negations ($P \wedge \sim q$, $\sim P \wedge q$, $P \wedge q$, $\sim P \wedge \sim q$) is called an elementary product.

↳ A logical expression is said to be DNF if it is sum of elementary products.

eg $P \vee (P \wedge \sim q)$
$P \vee (P \wedge q)$

Procedure to Obtain a DNF of a given logical expression —

1) Remove all $\leftrightarrow$ / $\rightarrow$ by an equivalent expression containing the connectives only. ($\wedge, \vee, \sim$).
2) Eleminate the negation ($\sim$) before sums & product by using $\sim\sim$ (double negation) or demorgan's law.
3) Apply the distributive law until the sum of elementary product is obtain.

Ques: $P \wedge (P \rightarrow Q)$

Soln: $P \wedge (\sim P \vee Q)$

$(P \wedge \sim P) \vee (P \wedge Q)$ DNF form. Ans.

Ques: $(P \rightarrow Q) \wedge \sim Q$

Soln: $(\sim P \vee Q) \wedge \sim Q$

$(\sim P \wedge \sim Q) \vee (Q \wedge \sim Q)$ DNF form. Ans

Ques: $\sim(P \wedge Q) \leftrightarrow (P \vee Q)$

Soln: $(P \wedge Q) \vee (\sim P \wedge \sim Q)$

$\Rightarrow [\sim(P \wedge Q) \wedge (P \vee Q)] \vee [\sim(\sim P \wedge Q) \wedge \sim(P \vee Q)]$

$\Rightarrow [(\sim P \vee \sim Q) \wedge (P \vee Q)] \vee [(P \wedge Q) \wedge (\sim P \wedge \sim Q)]$

$\Rightarrow \underbrace{[(\sim P \wedge P) \vee (\sim P \wedge Q)]}_{F} \vee \underbrace{[(\sim Q \wedge P) \vee (\sim Q \wedge Q)]}_{F} \vee \underbrace{[(P \wedge Q) \wedge (\sim P \wedge \sim Q)]}_{F}$

$(\sim P \wedge Q) \vee (\sim Q \wedge P)$. DNF form. Ans

Ques: $(P \rightarrow Q) \wedge (\sim P \wedge Q)$.

Soln: $(\sim P \vee Q) \wedge (\sim P \wedge Q)$.

$(\sim P \wedge \sim P) \vee (\sim P \wedge Q) \vee (Q \wedge \sim P) \vee (Q \wedge Q)$ DNF

Ans.

Ques: $(P \rightarrow Q) \rightarrow (\sim R \wedge Q)$

Soln: $(\sim P \vee Q) \rightarrow (\sim R \wedge Q)$

$[\sim(\sim P \vee Q) \vee (\sim R \wedge Q)]$

$(P \wedge \sim Q) \vee (\sim R \wedge Q)$ DNF Ans

CNF :- A logical expression is said to be CNF if it consist of product of elementary Sums.

Ques: $P \wedge (P \rightarrow Q)$

Sol: $P \wedge (\sim P \vee Q) \rightarrow$ CNF.

$(P \vee P) \wedge (\sim P \vee Q)$ Ans

Ques: $(P \rightarrow Q) \wedge \sim Q$

Sol: $(\sim P \vee Q) \wedge \sim Q \rightarrow$ CNF. Ans

Ques: $\sim(P \wedge Q) \leftrightarrow (P \vee Q)$

Sol: $(\sim P \wedge Q) \wedge (P \vee \sim Q)$

$[\sim(\sim P \wedge Q) \vee (P \vee Q)] \wedge [(\sim(P \wedge Q)) \vee \sim(P \vee Q)]$

$[(P \wedge Q) \vee (P \vee Q)] \wedge [(P \wedge Q) \vee (\sim P \vee \sim Q)] \rightarrow$ CNF.

Ans

$(\sim P \vee Q) \wedge (P \vee \sim Q)$

$[\sim(\sim P \wedge Q) \vee (P \vee Q)] \wedge [\sim(P \wedge Q) \vee \sim(P \vee Q)]$

$[P \vee \sim Q \vee P \vee Q] \wedge [\sim P \vee \sim Q \vee \sim P \wedge \sim Q]$

$[(P \vee (P \vee Q)) \wedge (\sim Q \vee (P \vee Q))] \wedge [(\sim P \vee (\sim P \vee \sim Q)) \wedge (\sim Q \vee (\sim P \vee \sim Q))]$

$[(P \vee Q) \wedge (\sim Q \vee P)] \wedge [(\sim P \vee \sim Q) \wedge (\sim Q \vee \sim P)]$

$[(P \vee Q) \wedge (\sim P \vee \sim Q)] \rightarrow$ CNF Ans

Ques: $(P \rightarrow Q) \wedge (\sim P \wedge Q)$

Sol: $(\sim P \vee Q) \wedge (\sim P \wedge Q) \Rightarrow (\sim P \vee Q) \wedge \sim(P \vee \sim Q)$ ✓

CNF.

$[(\sim P \wedge (\sim P \wedge Q)) \vee (Q \wedge (\sim P \wedge Q))]$

$[(\sim P \wedge \sim P)(\sim P \wedge Q) \vee (Q \wedge \sim P)(Q \wedge Q)]$ T

$[\sim(P \vee P)(\sim P \wedge Q) \vee (Q \wedge \sim P)]$ T

$(\sim P \wedge Q) \vee (Q \wedge \sim P)$

$[\sim(P \vee \sim Q) \vee \sim(\sim Q \vee P)]$

$\sim[(P \vee \sim Q) \wedge (\sim Q \vee P)] \rightarrow$ CNF. X Ans

Ques: $(P \to Q) \wedge (Q \vee (P \wedge R))$

Sol: $\sim(\sim P \vee Q) \wedge (Q \vee (P \wedge R))$

$\sim(\sim P \vee Q) \wedge ((Q \vee P) \wedge (Q \vee R))$ → CNF. Ans

Pdnf (Principal Disjunctive Normal form) :-

Let P & Q be the 2 statement variables, then, $P \wedge Q$, $\sim P \wedge \sim Q$, $\sim P \wedge Q$, $P \wedge \sim Q$ are called minterms of P & Q. It may be noted that none of the minterm should contain both the variable & it's negation.

Pdnf can be defined as an equivalent formula consisting of disjunctive of minterms only. This is also called the sum of product canonical form.

(minterm) ∨ (minterm) ∨ ----

T ∨ T ∨ ----

Ques: Obtain the Pdnf of a) $P \to Q$

b) $\sim P \vee Q$

c) $Q \vee (P \vee \sim Q)$

d) $(P \wedge \sim Q \wedge \sim R) \vee (Q \wedge R)$

Sol: a) $P \to Q$.

P	Q	$P \to Q$
T	T	T
T	F	F
F	T	T
F	F	T

Pdnf → $(P \wedge Q) \vee (\sim P \wedge Q) \vee (\sim P \wedge \sim Q)$ Ans

b)

p	q	~p	~p ∨ q
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

The Pdnf of ~p∨q =

(p∧q) ∨ (~p∧q) ∨ (~p∧~q)

c)

p	q	~q	p∨~q	q∨(p∨~q)
T	T	F	T	T
T	F	T	T	T
F	T	F	F	T
F	F	T	T	T

The Pdnf of q∨(p∨~q)

⇒

(p∧q) ∨ (p∧~q) ∨ (~p∧q) ∨ (~p∧~q)

d)

p	q	r	~q	~r	q∧r	p∧~q∧~r	(p∧~q∧~r)∨(q∧r)
T	T	T	F	F	T	F	T
T	T	F	F	T	F	F	F
T	F	T	T	F	F	F	F
T	F	F	T	T	F	T	T
F	T	T	F	F	T	F	T
F	T	F	F	T	F	F	F
F	F	T	T	F	F	F	F
F	F	F	T	T	F	F	F

The Pdnf of (p∧~q∧~r) ∨ (q∧r) =

(p∧q∧r) ∨ (p∧~q∧~r) ∨ (~p∧q∧r) Ans

Pcnf (Principal Conjunctive Normal Form):-

The dual of minterm is maxterm. For a given no. of variables the maxterm consist of disjunctions in

which each variable or its negation but not both, appears only once
(maxterm) ∧ (maxterm) ∧ - - - - -
(P∨Q), (~P∨Q), (P∨~Q), (~P∨~Q)
Pcnf can be defined as an equivalent formula consist of conjunctive of maxterm only. This is also the product of sums canonical form.

Ques Obtain the Pcnf of a) P∧Q
b) (~P→R) ∧ (Q↔P)
c) (P∧Q

Sol^n a)

P	Q	P∧Q
T	T	T
T	F	F
F	T	F
F	F	F

The Pcnf of P∧Q is = (~P∨Q) ∧ (P∨~Q) ∧ (P∨Q).

b)

P	Q	R	~P	~P→R	Q↔P	(~P→R) ∧ (Q↔P)
T	T	T	F	T	T	T
T	T	F	F	T	T	T
T	F	T	F	T	F	F
T	F	F	F	T	F	F
F	T	T	T	T	F	F
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	F	T	F

The Pcnf of (~P→R) ∧ (Q↔P) =
(~P∨Q∨~R) ∧ (~P∨Q∨R) ∧ (P∨~Q∨~R) ∧ (P∨~Q∨R) ∧ (P∨Q∨R)
~~(~P∨Q∨R.~~

Ans

Procedure for Obtaining a PDNF / PCNF :-

1. Obtain the CNF / DNF form.
2. Introduce missing factors.
3. Avoid duplication.

Ques Obtain the PDNF of the following:

(i) $P \vee \sim Q$

Sol: $[P \wedge (Q \vee \sim Q)] \vee [\sim Q \wedge (P \vee \sim P)]$

$\Rightarrow (P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim Q \wedge P) \vee (\sim Q \wedge \sim P)$

$\Rightarrow (P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$ PDNF Ans

(ii) $P \rightarrow Q$

Sol: $\sim P \vee Q$

$[\sim P \wedge (Q \vee \sim Q)] \vee [Q \wedge (P \vee \sim P)]$

$\Rightarrow (\sim P \wedge Q) \vee (\sim P \wedge \sim Q) \vee (Q \wedge P) \vee (Q \wedge \sim P)$

$\Rightarrow (\sim P \wedge Q) \vee (\sim P \wedge \sim Q) \vee (P \wedge Q)$. PDNF Ans

(iii) $Q \vee (P \vee \sim Q)$

Sol: $[Q \wedge (P \vee \sim P)] \vee [P \wedge (Q \vee \sim Q) \vee \sim Q \wedge (P \vee \sim P)]$

$(Q \wedge P) \vee (Q \wedge \sim P) \vee (P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim Q \wedge P) \vee (\sim Q \wedge \sim P)$

$\Rightarrow (P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$. PDNF Ans

(iv) $(P \wedge Q) \vee (\sim P \wedge R) \vee (Q \wedge R)$

Sol: $[P \wedge Q \wedge (R \vee \sim R)] \vee [\sim P \wedge R \wedge (Q \vee \sim Q)] \vee [Q \wedge R \wedge (P \vee \sim P)]$

$\Rightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \sim R) \vee (\sim P \wedge Q \wedge R) \vee (\sim P \wedge \sim Q \wedge R) \vee (P \wedge Q \wedge R) \vee (\sim P \wedge Q \wedge R)$.

$\Rightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \sim R) \vee (\sim P \wedge Q \wedge R) \vee (\sim P \wedge \sim Q \wedge R)$. PDNF Ans

Ques: Obtain the PCNF of the following:

(i) $P \wedge \sim Q$.

Soln: $[P \vee (Q \wedge \sim Q)] \wedge [\sim Q \vee (P \wedge \sim P)]$

$\Rightarrow [P \vee Q) \wedge (P \vee \sim Q) \wedge (\sim Q \vee P) \wedge (\sim Q \vee \sim P)$

$\Rightarrow (P \vee Q) \wedge (P \vee \sim Q) \wedge (\sim P \vee \sim Q)$. PCNF Ans

(ii) $P \leftrightarrow Q$

Soln: $(P \rightarrow Q) \wedge (Q \rightarrow P)$

$(\sim P \vee Q) \wedge (\sim Q \vee P)$

$[\sim P \vee (Q \wedge \sim Q)$ ~~$\wedge$~~ $Q \vee (P \wedge \sim P)] \wedge [\sim Q \vee (P \wedge \sim P) \vee P \vee (Q \wedge \sim Q)]$

$\Rightarrow (\sim P \vee Q) \wedge (\sim P \vee \sim Q)$ ~~$\wedge$~~ $(Q \vee P) \wedge (Q \vee \sim P) \wedge (\sim Q \vee P) \wedge (\sim Q \vee \sim P) \wedge$
$(P \vee Q) \wedge (P \vee \sim Q)$.

(iii) $P \wedge Q$.

Soln: $[P \vee (Q \wedge \sim Q)] \wedge [Q \vee (P \wedge \sim P)]$

$(P \vee Q) \wedge (P \vee \sim Q) \wedge (Q \vee P) \wedge (Q \vee \sim P)$

$\Rightarrow (P \vee Q) \wedge (P \vee \sim Q) \wedge (\sim P \vee Q)$. PCNF Ans

(iv) $(\sim P \rightarrow R) \wedge (Q \leftrightarrow P)$.

Soln: (~

Rules of Inference :- It is the criteria for determining the validity of an argument. Any conclusion which is arrived by following these rules is called And the argument is called a valid argument.

Argument :- An argument is the process by which conclusion is drawn from a sets of proposition. The given set of proposition is called "Premises" or "hypothesis". The final proposition derived from the given proposition is called "conclusion.

P_1
P_2
P_3
⋮
P_n } Premises / hypothesis
C ⇒ Conclusion

Agument is said to be a valid argument if and only if conjuction of all the premises implies the conclusion $(P \rightarrow C)$ i.e, if all the premises are true then the conclusion must also be true.

* $P_1 \wedge P_2 \wedge P_3 \wedge \cdots P_n \longrightarrow C \rightarrow$ Tautology
* $(P \rightarrow q) \wedge P \longrightarrow q$.

★ Imp

Qus: I study hard, then I get A's
I study hard

Soln: I get A's

let P → I study hard
Q → I get A's

$P \to Q$
P
$\therefore Q$

$(P \to Q) \wedge P \to Q$

This statement is valid through "modus Ponens."

P	Q	$P \to Q$	$(P \to Q) \wedge P$	$(P \to Q) \wedge P \to Q$	
T	T	T	T	T	} Tautology
T	F	F	F	T	
F	T	T	F	T	
F	F	T	F	T	

Rules of Inference — Name

(1) a) $\dfrac{P}{\therefore P \vee Q}$ b) $\dfrac{Q}{\therefore P \vee Q}$ — Addition

(2) a) $\dfrac{P \wedge Q}{\therefore P}$ b) $\dfrac{P \wedge Q}{Q}$ — Simplification

(3) $\dfrac{P}{Q}$
$\therefore P \wedge Q$ — Conjunction

(4) $p \to q$
p
$\therefore q$
Modus Ponens

(5) $p \to q$
$\sim q$
$\therefore \sim p$
Modus Tollens

(6) $p \vee q$
$\sim p$
$\therefore q$
Disjunction Syllogism

(7) $p \to q$
$q \to r$
$\therefore p \to r$
Hypothetical Syllogism

(8) $p \to q \wedge (r \to s)$
$p \vee r$
$\therefore q \vee s$
Constructive Dilemma

(9) $p \to q \wedge (r \to s)$
$\sim q \vee \sim s$
$\sim p \vee \sim r$
Destructive Dilemma

Ques: Represent the argument —
If this no. is divisible by 6, then it is divisible by 3.
This no. is not divisible by 3, therefore this no. is not divisible by 6.
Symbolically represent the argument.

Soln: P: This no. is divisible by 6.
Q: This no. is divisible by 3.

$$\begin{array}{l} P \rightarrow Q \\ \sim Q \\ \hline \therefore \sim P \end{array}$$

Hence it's validity by Modus Tollens.

Ques: If RAM works hard, he will get a job.
RAM works hard therefore he will get a job.
check validity.

Soln: P: Ram works hard.
Q: He will get a job

$$\begin{array}{l} P \rightarrow Q \\ P \\ \hline \therefore Q \end{array}$$

Hence it's valid by Modus Ponens.

Ans

Ques: If a Man is Bachelor he is an unhappy.
If a man is unhapply he dies young.
Therefore bachelor's die young.
Check validity.

Soln: P: Man is Bachelor
Q: He is unhappy.
R: Bachelor's die young.

$$\begin{array}{l} P \rightarrow Q \\ Q \rightarrow R \\ \hline \therefore P \rightarrow R \end{array}$$

Hence it's valid by Hypothetical Syllogism.

Ans

Ques: Represent the argument –
Either Ram is not guilty or Shyam is telling a truth.
Shyam is not telling a truth therefore Ram is not guilty.
Symbolically represent & determine it is valid.

Sol: q: Ram is not guilty
p: Shyam is telling a truth.
$\sim p$: Shyam is not telling a truth.

$p \vee q$
$\sim p$
$\therefore q$

Hence it's valid by Disjunction syllogism.

Ans

Ques: Show that it is a valid conclusion from the premises $p \to q$, $q \to r$, $r \to s$, $\sim s$ & $p \vee t$

Sol:
1. $p \to q$ (Premises)
2. $q \to r$ (Premises)
3. $r \to s$ (Premises)
4. $p \to r$ (Hypothetical syllogism by using 1 & 2)
5. $p \to s$ (Hypothetical syllogism by using 3 & 4)
6. $\sim s$ (premises)
7. $\sim p$ (Modus Tollens by using 5 & 6)
8. $p \vee t$ (Premises or apply by Disjunctive syllogism by using 7 & 8)
9. t (valid)

Ans

Hence valid.

Ques: S is a valid conclusion from the premises, $P \to Q$, $P \to R$, $\sim(Q \wedge R)$ & $S \vee P$.

Sol:

(I) $P \to Q$ (Premises)

(II) $P \to R$ (Premises)

(III) $(P \to Q) \wedge (P \to R)$ (By using 1 & 2) (Add)

(IV) $\sim(Q \wedge R)$ (Premises)

(V) $\sim Q \vee \sim R$ (By Demorgan's law)

(VI) $\sim P \vee \sim P = \sim P$ (By destructive dilemma using 3 & 5)

(VII) $S \vee P$ (Premises)

(VIII) S (By disjunctive syllogism using 6 & 7)

Hence valid.

Ans

Ques: Prove the validity of following argument.

"If I get the job and work hard then I will get promoted.

If I get promoted then I will be happy.

I will not be happy.

Therefore either I will not get the job or I will not work hard."

Sol:

P: I get the job.

Q: I will get work hard.

R: I get promoted.

S: I will be happy.

$(P \wedge Q) \to R$ (Premises)

$R \to S$ (Premises)

$\sim S$ (Premises)

$\therefore \sim P \vee \sim Q$

(I) $(P \wedge Q) \rightarrow R$ (Premises)

(II) $R \rightarrow S$ (Premises)

(III) $\sim S$

(IV) $(P \wedge Q) \rightarrow S$ (Hypothetical syllogism by using 1 & 2)

(V) $\sim (P \wedge Q)$ (By Modus tollens by using 3 & 4)

(VI) $\sim P \vee \sim Q$ (By Demorgan's law)

Hence valid. Ans.

Ques: Establish the validity of the following by conditional proof.

$P \wedge ((P \rightarrow \sim Q) \vee (R \wedge S)) \rightarrow (Q \rightarrow R)$

Ques Determine the validity of statement —
If $7 < 4$ then 7 is not prime no.
7 is not less than 4 therefore 7 is prime no

Soln $P : 7 < 4$

$\sim P : 7 \not< 4$

Q : 7 is prime no.

$\sim Q$: 7 is not prime no.

$$\begin{array}{l} P \rightarrow \sim Q \\ \underline{\sim P} \\ \therefore Q \end{array}$$

P	$\sim P$	Q	$\sim Q$	$P \rightarrow \sim Q$	$P \rightarrow \sim Q \wedge \sim P$	$P \rightarrow \sim Q \wedge \sim P \rightarrow Q$
T	F	T	F	F	F	F
T	F	F	T	T	F	T
F	T	T	F	T	F	T
F	T	F	T	T	F	F

Hence, it is not tautology. Ans
not valid

Ques: Is the validity :
If two sides of triangle are equal then opposite angles are ~~triangle~~ equal.
Two sides of triangle is not equal.
∴ Opposite angles are not equal.

Soln P: Two sides of triangle are equal.
Q: Opposite angles are equal.

★ $\forall x$ → Universal (→) if - - then

★ $\exists x$ → Existential (∧) and

Date:
Page:

$P \rightarrow Q$

$\sim P$

$\therefore \sim Q$

P	~P	Q	~Q	P→Q	(P→Q)∧~P	P→Q∧~P→~Q
T	F	T	F	T	F	T
T	F	F	T	F	F	T
F	T	T	F	T	T	F
F	T	F	T	T	T	T

not tautology. Hence invalid. Ans.

Limitation of Propostional logic :- We can't represent the words like, every, each, none, or, someone, none of these etc with the help of propositional logic. like,

eg:- All boys are intelligent
Some apples are sweet.
propositional logic.

→ This type of sentence can be present in propositi-onal logic but not suitable. These statement can be written as with help of predicate logic.

$\forall x$: student(x)
predicate.

↳ In propositional logic we can not describe state-ment of there properties or logical relationship.

P_1 : Every student is brilliant.
P_2 : Riyansh is a student.
P_3 : Riyansh is brilliant.

↳ Above argument can not be represented with the propositional logic but it can be represent with the help of predicate logic.

Predicate logic (FOPL) / (FOL)

* First Order Predicate Logic (FOPL)
* First Order logic (FOL)

It is a statement involving the statement

eg: x is an animal.
We can not say these statement true or false untill we do not specify the value of x & y.

x is an animal. → can be represented
Subject ↑ Predicate by P(x)
where P() is predicate & x is subject / variable.

After assigning the value of x either true / false

P(x) : x is an animal.
P(dog) : dog is an animal - True.
P(man) : man is an animal - False.

eg: (1) Ramesh is a student
student (Ramesh)

(2) Dog is an animal.
animal (dog)

Ques: P(x) denote the statement $x > 3$. what are the truth value of P(4), (2)

Soln: P(4) - True $\{x > 3 \Rightarrow 4 > 3\}$
P(2) - false $\{x > 3 \Rightarrow x < 3\}$

Ques: let P(x) denote the statement word contain letter "a". what are the truth value P(orange), P(True), P(false)

Soln: P(orange) {P contains "a"} - True
P(True) {P does not contains "a"} - false
P(false) {P contains "a"} - True

Ques: P(x, y) denote "x is a capital of y". What is the truth value of (1) P(Lucknow, India) - false
(2) P(Bhopal, M.P.) - True
(3) P(Patna, Bihar) - True
(4) P(Agakla, Meghalay) - false
(5) P(Guvahati, Assam) - false

Quantifiers :- Quantifiers are the words that refers to the quantities such as some, all, few, many, none, etc. It indicates about how ~~many~~ ~~elements~~ frequently a certain statement is true.

* Types of Quantifiers :-

1) Universal Quantifiers :- It is represented by "$\forall_x$"
Universal Quantifiers of $P(x)$ is a statement "$P(x)$ $\forall$ value of x is domain is True.

$\forall_x . (P(x))$: for all x, $P(x)$

eg for all, every, each, all, any, etc.

Ques All human beings are Mortal.

Sol^n $P(x)$: x is mortal.
x means human beings.

OR

$\forall_x P(x) \longrightarrow$ Mortal. OR $(\forall_x \in U) P(x)$.
All human beings

Ques Every Person is Precious.

Sol^n For every x, If x is a person then x is precious.
$A(x)$ $B(x)$

$A(x)$: x is person.
$B(x)$: x is precious

$\forall_x A(x) \longrightarrow B(x)$ or $\forall_x$ Person$(x) \longrightarrow$ Precious(x).

Ques All students are good.

Sol^n For all x, If x is a student then x are good.
$A(x)$ $B(x)$

$\forall_x A(x) \longrightarrow B(x)$.

Ques: All boys likes football.

Sol^n: For all x, if x are boys then x likes football.

$\forall x\, A(x) \longrightarrow B(x)$ or $\forall x$ boys$(x) \longrightarrow$ likes$(x, \text{football})$

Ques: All bad students were commenting on girls.

Sol^n: For all x, if x is bad student then x were commenting on girls.

Ques: All birds can fly.

Sol^n: for all x, if x is bird then x can fly.

$\forall x$ birds$(x) \longrightarrow$ fly(x)

Ques: All men are genious.

Sol^n: for all x, if x is men then x are genious.

$\forall x$ men$(x) \longrightarrow$ genious(x)

Ques: Not all students like math.

Sol^n: $\neg \forall x$ students$(x) \longrightarrow L(x)$ (math)

Ques: All student play cricket.

Sol^n:

Ques: All pompeian were not Roman.

Sol^n:

Ques: All students need financial aid.

Soln

Ques: Every Gardner likes Sun.

Soln

Ques: Not all students likes both history & geography.

Soln $\neg \forall x \; M(x) \longrightarrow L(x)\,(\text{History} \wedge \text{geography})$.

OR

$\neg \forall x \; \text{Students}(x) \longrightarrow [\text{likes}(x, \text{history}) \wedge \text{likes}(x, \text{geography})]$.

Existential Quantifier:- According to this quantifier there is one or more variables in the particular domain is true, but not all (some, atleast, only one, atleast one, etc).

The phrase their exist denoted by "$\exists$" is called existensial Quantifier.

eg: "There exist x such that $x^2 = 5$"

$(\exists x \in R)\, P(x)$ OR

$\exists x \; P(x)$

eg: Some formulas are tautology

$F(x) = x$ is formula.

$T(x) = x$ is Tautology

$\exists x : F(x) \wedge T(x)$

eg: Some boys like football.

$B(x)$: x is a boys

$f(x)$: x likes (football)

$\exists x : B(x) \wedge f(x)$

eg: Some no.'s are not rational.

$N(x)$: x is a no.

$R(x)$: x is rational

$\exists x : N(x) \wedge \sim R(x)$

eg: Some cows are not white.

$C(x)$: x is cows

$W(x)$: x are white

$\exists x : C(x) \wedge \sim W(x)$

eg: Some real no.'s are rational

$R(x)$: x is real no.

$r(x)$: x are rational

$\exists x : R(x) \wedge \sim r(x)$

eg: There doesnot exist any person who is my friend and perfect.

eg: There are some friends who are not perfect.

$F(x)$: x is a friends.

$P(x)$: x are perfect

$\exists x : F(x) \wedge \sim P(x)$.

eg: Some people who are not my friend are not perfect.

$F(x)$: x are friends.

$P(x)$: x are perfect.

$\exists x : \sim F(x) \wedge \sim P(x)$

eg: Some people who are not my friend are perfect.

$F(x)$: x is a my friend

$P(x)$: x is a perfect

$\exists x : \sim F(x) \wedge P(x)$

eg: Every lion likes alive meat.

eg: Only one student fail in math

$S(x)$: x is a students

$M(x)$: x fail in math.

$\exists x : S(x) \wedge M(x)$

Statement	Negation
all true $\forall x\, P(x)$	$\exists x [\sim P(x)]$ at least one false
at least one false $\exists x [\sim P(x)]$	$\forall x\, P(x)$ all true
all false $\forall x [\sim P(x)]$	$\exists x\, P(x)$ at least one true
at least one true $\exists x\, P(x)$	$\forall x [\sim P(x)]$ all false.

Ques 20

DNF :- $P \rightarrow (P \wedge (q \rightarrow P))$

$P \rightarrow (P \wedge (q \rightarrow P))$

$\sim P \vee (P \wedge (q \rightarrow P))$

$\sim P \vee (P \wedge (\sim q \vee P))$

$\sim P \vee [(P \wedge \sim q) \vee (P \wedge P)]$

$\sim P \vee (P \wedge \sim q) \vee (P \wedge P)$

$(P \wedge \sim P) \vee (P \wedge \sim q) \vee (P \wedge P)$ DNF. Ans

Ques 20

CNF :- $\sim (P \vee q) \leftrightarrow (P \wedge q)$

(P = $P \vee q$, q = $P \wedge q$)

$P \leftrightarrow q = (\sim P \vee q) \wedge (P \vee \sim q)$

$[(P \vee q) \vee (P \wedge q)] \wedge [(\sim P \wedge \sim q) \vee (P \wedge q)]$

$[(P \vee q) \vee (P \wedge q)] \wedge [(\sim P \wedge \sim q) \vee (\sim P \vee \sim q)]$

$[(P \vee q \vee P) \wedge (P \vee q \vee q)] \wedge [(\sim P \vee \sim q \vee \sim P) \wedge (\sim P \vee \sim q \vee \sim q)]$

$(P \vee q) \wedge (P \vee q) \wedge (\sim P \vee \sim q) \wedge (\sim P \vee \sim q)$

$(P \vee q) \wedge (\sim P \vee \sim q)$. CNF. Ans